

D Eeshan Pavan Karteek

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Professional Summary

Computer Science (AI & ML) undergraduate with experience in Machine Learning, Backend Development, and Distributed Systems. Built scalable applications using Python, Java, Node.js, React, TensorFlow, and Kafka. Developed real-time optimization systems, deep learning models, and multi-tenant web platforms with measurable improvements in performance, accuracy, and efficiency.

Education

Sri Venkateshwara College of Engineering

2022 – 2026

B.Tech in Computer Science and Engineering (AI & ML)

CGPA: **8.93/10**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Machine Learning, Artificial Intelligence, OOP, Software Engineering

Technical Skills

Programming Languages: Python, Java, C

Frontend: React, TypeScript, JavaScript, HTML, CSS, Vite

Backend: Node.js, Express.js, REST APIs, Flask

Machine Learning & AI: TensorFlow, Scikit-Learn, XGBoost, CNNs, Deep Learning, Generative AI, Gemini API

Databases: MySQL, PostgreSQL, Firebase, PostGIS, JSON Database

Tools & Technologies: Kafka, Git, GitHub, JWT, BcryptJS, OR-Tools, Pandas, Jupyter Notebook, Google Colab

AI & Generative AI: Prompt Engineering, Generative AI, Gemini API, LLM Applications, AI Agents

Experience

Gen AI Intern — Smart Bridge

Dec 2025 – Mar 2026

- Developed a Generative AI application using Gemini API supporting both text and image-based inputs.
- Engineered an end-to-end workflow for input processing, prompt generation, and response orchestration.
- Automated recipe generation and nutritional analysis, reducing manual search effort by approximately 70%.
- Implemented image understanding pipelines to extract food information and generate contextual recommendations.
- Validated system performance across 20+ user scenarios and test cases.

Projects

Last-Mile Logistics Optimizer

Python, OR-Tools, Kafka, XGBoost, PostGIS, TomTom API

- Built a real-time route optimization platform integrating live traffic and weather data.
- Reduced fuel costs by 20% and delivery time by 15% using traffic-aware route planning.
- Improved ETA prediction accuracy by 25% through ML-based dwell-time estimation.
- Enabled sub-second route re-optimization using Kafka-driven event processing.
- Increased driver capacity by 10% through dynamic route adjustments.
- Evaluated performance across 300+ simulated delivery routes using real-world traffic datasets.

Multi-Tenant Ticketing System

React, TypeScript, Node.js, Express.js

- Developed a secure multi-tenant ticketing platform with Organization-Based Access Control (OBAC).
- Implemented a native SMTP listener to convert incoming emails into structured support tickets.
- Designed a race-condition-resistant backend using atomic JSON storage and sequential write queues.

- Built contextual search capabilities for efficient cross-tenant data retrieval.
- Enhanced security through JWT authentication, BcryptJS password hashing, and administrator protection workflows.

Bone Fracture Classification Model

TensorFlow, Flask, Scikit-Learn, XGBoost

- Engineered and benchmarked CNN architectures including MobileNet, VGG19, EfficientNet, and DenseNet.
- Improved classification accuracy from 84.2% to 92.4% using a hybrid DenseNet121 + XGBoost architecture.
- Trained and evaluated models on 10,000+ labeled X-ray images spanning 10 fracture classes.
- Deployed inference services using Flask APIs for real-time predictions.
- Evaluated performance using Precision, Recall, and F1-Score metrics.

Achievements

- Maintained a CGPA of 8.74 in B.Tech Computer Science (AI & ML).
- Built end-to-end AI, Machine Learning, and Full-Stack software projects using industry-standard technologies.
- Demonstrated expertise across backend development, distributed systems, and deep learning workflows.